Tutorial 6. Molecular Geometry and Bonding theory

Exercise 1. ***(Know)***

How many nonbonding electron pairs are there in each of the following molecules:

(a) $N(CH_3)_3$,

(b) CO,

(c) BF_3,

(d) SO_2

Exercise 2. ***(Und)***

The figure that follows shows ball-and-stick drawings of three possible shapes of an AF_3 molecule.



(i) (ii) (iii)

(a) For each shape, give the electron-domain geometry on which the molecular geometry is based.

(b) For each shape, how many nonbonding electron domains are there on atom A?

(c) Which of the following elements will lead to an AF_3 molecule with the shapes (i), (ii) and (iii): Li, B, N, Al, P, Cl?

Exercise 3. ***(Und)***

Use the VSEPR model to predict the molecular geometry of (**a**) CO_2, **(b)** H_2O, **(c)** SF_4, **(d)** IF_5

Exercise 4: ***(Und)***

Predict whether these molecules are polar or nonpolar: (a) BrCl, (b) SO_2, (c) SF_6

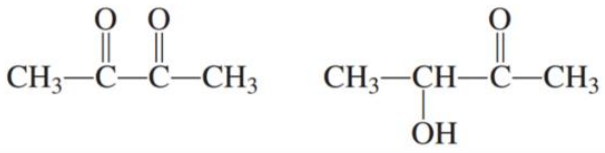
Exercise 5: ***(Und)***

Indicate the hybridization of the central atom in

a) BCl_3 b) $AlCl_4^-$ c) CS_2 d) GeH_4 e) $SiCl_4$ f) HCN g) SO_3 h) $TeCl_2$ i) SF_6 j) PCl_5

Exercise 6: ***(Know-Und)***

Biacetyl and acetoin are added to margarine to make it taste more like butter.



Biacetyl Acetoin

a) Complete the Lewis structures and Predict values for all C—C—O bond angles

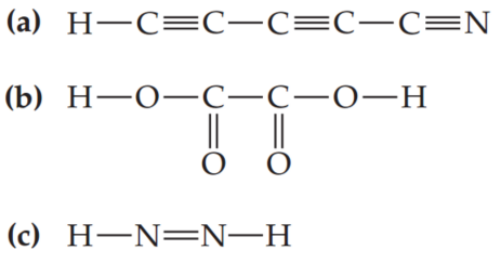
b) Give the hybridization of the carbon atoms in these two compounds

c) Must the four carbon atoms and two oxygen atoms in biacetyl lie on the same plane?

d) How many σ bonds and how many π bonds are there in biacetyl and acetoin.

Exercise 7 *(App)*

Predict the molecular geometry of each of the following molecules?



Exercise 8 *(App)*

a) Drawing the MO diagram of following species and arrange them in order of increasing stablitity.

He_2^{2+}, He_2^{+}, He_2

b) Using MO model, explain why:

(i) The peroxide ion O_2^{2-} has a longer bond length and weaker bond strength than the superoxide O_2^{-} and O_2^{2+}

(ii) The removal of one electron in O_2 strengthens bonding, while the removal of one electron in N_2 weakens bonding.